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RESEARCH ARTICLE

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Enhancing policy legitimacy for less and slower mobility: the potential of catalytic policy instruments

Claus Hedegaard Sørensen, Malin Henriksson and Jens Portinson Hylander

The Swedish Road and Transport Research Institute (VTI), Lund, Sweden

ABSTRACT

To avoid exceeding planetary boundaries, individual transport has to change, which, for many people, implies less and slower mobility. This contradicts the current hypermobility paradigm, making it difficult to gain legitimacy for such policies. This article contributes to conceptual development regarding how policies for less and slower mobility can gain legitimacy by developing the concept of *catalytic policy instruments*, i.e. policy instruments that can punctuate equilibria and transform policy agendas. Three such instruments are analysed through European examples that assist our theoretical reflections. We consider their respective potential and preconditions for being catalytic. The instruments are *visioning,* which may contribute to changing norms and values and thereby establish the legitimacy of policy preconditions; *public participation,* which may make policy procedures more legitimate; and *experimentation,* which may contribute to legitimising anticipated policy effects. We also identify directions for future research on catalytic policy instruments to improve the understanding of their applicability and limitations for policy making. While not being Columbus's egg, the strategic and tactical use of catalytic policy instruments is important for building a counterhegemonic alternative to the unsustainable hypermobility society.

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Less and slower mobility; policy legitimacy; catalytic policy instruments; public participation; visioning; experimentation

1 Introduction

The need for rapid and large-scale transformations to avoid exceeding planetary boundaries is reflected in international policy, including sustainable development goals (United Nations, 2015a), the Paris Agreement on Climate Change (United Nations, 2015b), and the Convention on Biological Diversity (United Nations Environmental Programme, 2011). This is especially the case for the transport sector, from which fossil fuel emissions have continued to increase, despite gradual improvements in many other sectors (European Environment Agency, 2025). Accordingly, the Intergovernmental Panel on Climate Change (IPCC) states that decarbonisation of the transport sector requires a '*paradigm shift* that ensures prioritisation of high-accessibility transport solutions that *minimise the amount of mobility* required to meet people's needs and *favours transit and active transport modes*' (IPCC 2022, 1057, italics added). This paradigm shift indicates a need to strengthen *avoid* and *shift* policies (Schipper & Marie-Lilliu, 1999; United Nations, 2016), simultaneously reducing overall travel needs and changing travel patterns away from automobility and aeromobility towards public transport and other shared modes, as well as active travel modes (Banister, 2008; Gudmundsson et al., 1996).

Since existing mobility options are deeply unequally distributed (Martens, 2017; Sheller et al., 2020), a shift in policy would need to strengthen the conditions for using public transport and other active modes, which would result in improved mobility options for many. However, for others, a consequence would be *less and slower mobility* compared to the current situation. Because the length and, possibly, number of trips are reduced (von Schönfeld & Ferreira, 2022) and mobility is slower, trains in many connections take more time than air transport does and because public transport, other shared modes, and biking and

CONTACT Claus Hedegaard Sørensen claus.h.sorensen@vti.se The Swedish Road and Transport Research Institute (VTI), Bruksgatan 8, 222 36 Lund, Sweden

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walking outside of the centre of cities usually cannot compete with the car in speed (Liao et al., 2020; Saunier & Chabin, 2020).

In light of this, a critical policy question is how to improve legitimacy for policies that promote less and slower mobility. To answer this gap, the aim of this article is to contribute to conceptual development regarding policies to gain legitimacy for less and slower mobility. We do this by presenting and discussing examples of policy instruments that can be suited to build legitimacy for policies favouring less and slower mobility. We introduce the term *catalytic policy instruments*, i.e. policies that can catalyse transformation, to characterise this type of instrument. Drawing on examples from previous studies indicates that the topic has—at least to some extent—been dealt with in the transport literature; however, it usually does not have a comprehensive focus on several catalytic policy instruments and their interlinkages. The research questions we aim to answer are as follows:

- What characterizes catalytic policy instruments?
- In what way can catalytic policy instruments contribute to the establishment of conditions that favour policies of less and slower mobility?

The article proceeds in four sections. In Section 2, the core theoretical concepts are introduced. Section 3 focuses on methodological considerations. In Section 4, we discuss envisioning exercises, public participation, and experimentation as three examples of potentially catalytic policy instruments for less and slower mobility and relate them to empirical research. In Section 5, we discuss our findings and finally draw conclusions, including what kind of further research is needed on catalytic policy instruments, in Section 6.

2 Legitimacy, policy legitimacy and catalytic policy instruments

To ground the notion of catalytic policy instruments for less and slower mobility, in this section, we discuss the theoretical foundations for the advancement of catalytic policy instruments for less and slower mobility by relating it to the theoretical literature on legitimacy and policy legitimacy.

2.1 Legitimacy and policy legitimacy

Policies for less and slower mobility fundamentally contradict the high mobility (Kaufmann et al., 2015) or hypermobility (Cohen et al., 2015) society that affluent populations are accustomed to. A high-mobility society offers fast, cheap and comfortable transport options centred around private cars and airplanes. It can be anticipated that large portions of the highly mobile global population at first glance would find a future of less and slower mobility unattractive, not least since the 'glamorization of mobility, and in particular hypermobility, now involves a wide range of elements and dimensions' (Cohen et al., 2015, 1667), and is underpinned by a range of discourses that people with high-carbon lifestyles (including excessive car-driving and frequent flying) deploy to justify and normalise their consumption patterns (Cass et al., 2023).

While there are groups that question the fast mobility options and long travelling distances, (e.g. the slow city movement, Cittaslow[1] or We Stay On the Ground[2]) and broad popular support for climate action exists (André et al., 2024), protests often arise when climate goals develop into concrete policy measures (Wanvik & Haarstad, 2021; Portinson Hylander et al., 2024), indicating a policy backlash to climate policies (Patterson, 2023). To implement policies that favour less and slower mobility, legitimacy for them needs to be achieved. At its core, legitimacy concerns how governing and the exercise of power are justified and how trust in systems is upheld. Legitimacy has been categorized analytically in diverse ways, from Max Weber's sociological distinction between *legal*, *traditional* and *charismatic* legitimacy (Steffek, 2019), through Beetham's political-philosophical terms *conformity to rules*, *justification by reference to shared beliefs*, and *consent by the subordinate* (Beetham, 2013), to the political science notions of *input* and *output* legitimacy (Scharpf, 1999), to which *throughput* legitimacy has later been added (Schmidt, 2013). Common to all conceptualizations are attempts to identify the specific aspects of regimes, perceptions or processes that lend legitimacy to a specific form of governing.

In legitimacy studies, research has suggested that '[w]e are far more familiar with legitimacy as a need of states or regimes' (Smoke, 1994, 98) and that 'single policies have [not] been a main interest of legitimacy studies' (Matti, 2009, 43). Furthermore, when single policies have been studied, the purpose often has been to study the effect on the regime in question. There is, of course, a close connection between the legitimacy of a regime and its policies; a regime with higher legitimacy will likely find it easier to gain acceptance for its policies, and well-received policies will likely improve regime legitimacy. While acknowledging regime legitimacy as an important contextual factor, in this study, we focus on the legitimacy of specific policies.

Similar to the general legitimacy concept, a variety of subcategories have been suggested and applied in the policy legitimacy literature, e.g. input–output terminology (Jagers et al., 2016; Kronsell, 2013), procedural-substantive (Wallner, 2008) and normative and cognitive dimensions (George et al., 1980; Smoke, 1994). These definitions all denote different dimensions of policy-making that influence legitimacy and can be divided into three distinct aspects of the policy process: (i) the *preconditions* for policy making, i.e. to what extent policies align with existing conditions, norms and values (normative legitimacy); (ii) the *procedures* through which policies are developed and decided (input/procedural legitimacy) and (iii) the *effects* the policies have, or are expected to have, on a target population (substantive/output/cognitive legitimacy) (Figure 1). Previous studies have argued that there are both trade-offs and compensation between these aspects, meaning that one dimension of legitimacy may compensate for another and that a substantial lack of legitimacy in one area may hamper other dimensions (Kronsell, 2013). Furthermore, it is important to note that the contents of all subcategories are dynamic and will change; what is considered in accordance with attitudes, norms and values and what is considered legitimate processes will differ and change over time, both within and between different societies and regimes.

2.2 Catalytic policy instruments

To advance the understanding of policy legitimacy for less and slower mobility, we propose the concept of 'catalytic policy instruments'. Policy instruments, or their neighbouring concepts 'policy tools', 'tools of government' or 'governance tools' (Bali et al., 2021; Howlett, 2008), can be defined as 'the techniques of governing that help define and achieve policy goals' (Bali et al., 2021), i.e. the specific measures implemented to assist governments in reaching set goals for society, or parts of it. However, much research has shown that the choice of policy instruments is not a purely rational or instrumental process but is infused with ideology and fashions (Lascoumes & Le Gales, 2007); sometimes, the choice of instrument even precedes decisions on goals and aims (Kingdon, 2003).

Policy instruments can play different roles. Some are 'intended to directly affect the nature, types, quantities and distribution of goods and services provided in society' (Howlett, 2008, 415), while others are focused on managing state–societal interactions in order to assure general support for government aims and initiatives' (ibid. 412) and might affect *how* a policy is crafted, formulated and implemented (Bali et al., 2021). All types of policy instruments, whether they deal mostly with the nature and distribution of goods and services or procedures and processes, can prove important to gain legitimacy for policies of less and slower mobility.

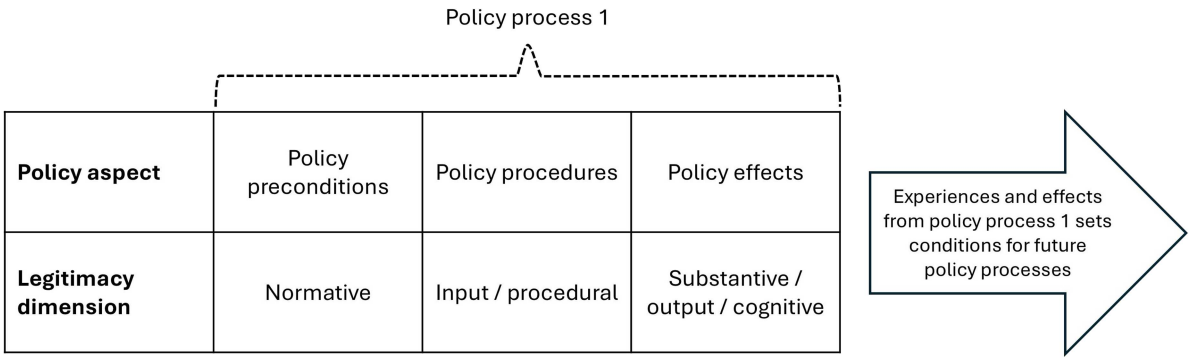


Figure 1. Theoretical model of relations between policy process aspects and legitimacy categories.

Catalytic policy instruments have the potential to substantially transform the conditions and shift the groundwork for future policy making. The term is inspired by the narrower term 'catalytic procedural tool', introduced by Stark & Yates (2021) in connection with an analysis of public enquiries as a tool that may transform policy agendas. 'Catalytic' refers to the ability of the instrument to 'punctuate equilibria and transform policy agendas' (ibid. 345) and thereby provide conditions for introducing other transformative policy instruments. The notion of a 'catalytic' policy instrument thus implies that it can substantially influence the policy process and change the legitimation of certain policies. Drawing on the previous discussion of different forms of legitimacy, we argue that the catalytic potential of certain policy instruments lies in shifting the level of a policy's legitimacy either in an ongoing policy process or in subsequent processes, since experiences from one policy process are carried out to consecutive ones. If a policy instrument succeeds in catalysing change, either the outcome of the first policy process or the point of departure for subsequent policy processes will be different than it would have been without the use of the policy instrument (see Figure 2).

The term 'catalytic policy instrument' is a generic term that can be employed to analyse any policy area. In this context, we are interested in understanding which policy instruments can improve legitimacy for less and slower mobility. This, thus, signifies policy instruments that, in one way or another, may provide public support for policies that counter hypermobility and support transitions toward lifestyles that require less travel. We write 'may', as it is not a given that one type of policy instrument is catalytic under any circumstance or in any context.

3 Methodological considerations

To analyse the potential of catalytic policy instruments for less and slower mobility, we focus on three different policy instruments: *visioning exercises, public participation*, and *experimentation*. These policy instruments each correspond to an aspect of the policy process: visioning the preconditions, public participation in the procedures and experimentation with the effects of policy making. Since our ambition is theoretical elaborations, we select illustrative cases that serve as a basis for theoretical reflections (Flyvbjerg, 2006). The examples have been chosen based on our previous experiences of doing research on public participation (cf. Sørensen et al., 2023; Sørensen et al., 2025; Portinson Hylander et al. submitted), visioning (cf. Henriksson et al. 2024a) and experimentation (cf. Henriksson et al., 2024b; Oldbury & Henriksson, submitted). The purpose of this theoretical exercise has been to situate the concept of catalytical policy instruments both in the general literature on legitimacy and policy instruments, as

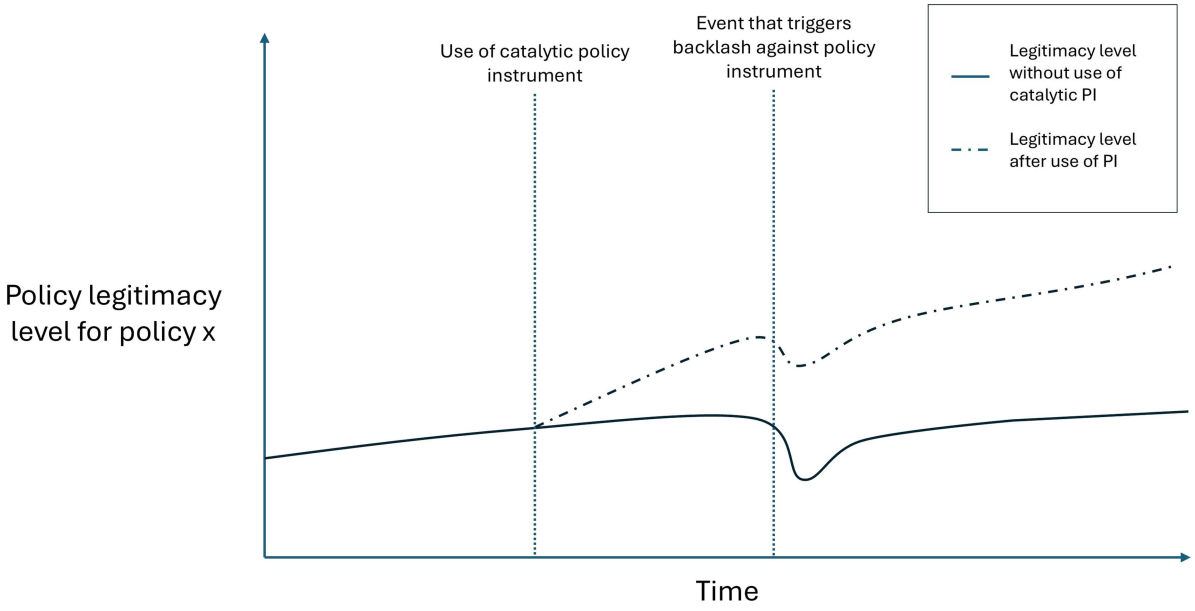


Figure 2. Theoretical model of how catalytic policy instruments may affect policy legitimacy.

discussed above, and in the literature on particular empirical topics, e.g. visions, public participation and experimentation (in Section 4).

4 Catalytic policy instruments: three examples

In this section, we present and elaborate on three examples of potential catalytic policy instruments for less and slower mobility. For each example, we consider the characteristics of the policy instrument in question, how it relates to legitimacy, and which aspect of the policy process it contributes to providing legitimacy: in terms of policy preconditions, policy procedures or policy effects.

4.1 Visioning

Visions function as a way for policy makers to influence norms regarding the direction society should develop, and the problems that should be addressed to do so (Syssner, 2018, building on Rothman, 2011). The deliberate use of visions in planning or political processes can be described as a policy instrument that aims to influence the norms of both those who govern and those who are governed, thus supporting legitimacy concerning the policy preconditions in particular. Visions can be a key instrument to attract the interest of policy makers and citizens in addressing pressing issues and committing to, e.g. climate targets (ibid.). Following this line of thought, some researchers and activists argue that positive visions can play a crucial role in guiding urban transformation and motivating actions among citizens (Isaksson et al., 2014; McPhearson et al., 2016; Skoglund, 2019). Hölscher et al., 2022 argue that visions can be transformative when they are *counterhegemonic*. This means that a vision must suggest a future that is radically different from the present state and, in doing so, explicitly challenges and questions current norms.

4.1.1 Visions for less and slower mobility

A strand of research focuses on how visions of the future can support sustainability policy in the transport sector (see Henriksson et al., 2024a). In the following, we highlight insights that can be drawn from visioning exercises that seek to formulate positive visions of transport futures and that could support legitimacy for reduced and slower mobility.

The first example is an EU project that developed four scenarios for mobility in Europe in 2030 (Keseru et al., 2021). A research team used a scenario workshop methodology in which researchers draft scenarios based on current knowledge, including a survey of current trends that are likely to affect the future of transport, and discuss and develop scenarios with experts and citizens in workshops to reach a consensus on the most preferred future. After this process, a visioning workshop with stakeholders set a common vision for the future. The stakeholder group included experts from different areas within transport (policy makers, land use planners, vehicle manufacturers and providers, freight experts, etc.) and citizens representing young people, disabled and older transport users, public transport users, pedestrians, and cyclists. Four normative scenarios with different degrees of technological innovation, digitalization and support of local active transport laid the groundwork for the envisioning workshops. A majority of the stakeholders supported a scenario where governments are pressured to change the behaviour of citizens and companies and reduce carbon emissions to respond to climate change. Reducing travel demands and focusing on accessibility and services at the local level can support people in living slower, healthier, and more active lives. In the vision process, cyclists and pedestrians tend to prefer scenarios that support 'slow modes'. In the most preferred scenarios, governments play a crucial role in funding and supporting sustainable transport, with strict regulations to limit CO_2 emissions or to support digital innovation. Keseru et al. (2021) argue that their approach to building future visions leads to a consensual and desirable future pathway that has the support of a wide range of stakeholders. We also note that there was little support for a hypermobile future where more and faster travel continues to be normalized; rather, stakeholders supported a wish for local lifestyles.

The second example is the coproduction of four transformative visions that aim to address climate change and exploit opportunities for sustainability and resilience at different scales in Europe (Hungary, Scotland, Iberia, Europe) (Hölscher et al., 2022). The visions were developed collaboratively by various stakeholders within an EU-funded initiative with the aim of developing a common sustainable vision for

Europe that can be adopted in different national and regional contexts. The stakeholders were 'forerunners' and engaged in sustainability issues in diverse ways. The visions demonstrate shared ambitions for a low-carbon and climate-resilient Europe and thus encompass issues beyond transport and mobility but also included this theme specifically. In particular, the Hungarian vision emphasizes the need for slower mobility in terms of a focus on walking and cycling, while all the visions support different forms of electric and virtual mobility. To formulate a transformative vision, this example highlights that mitigation and climate goals need to be combined with goals for wellbeing, justice and social as well as sustainable development. Therefore, all four visions emphasize high quality of life, community life, and equality among citizens and societies. This was made possible by not only introducing a planetary and social boundaries framework but also playful and artistic elements to spark transformative thinking in the visioning process. Hölscher et al. (2022) argue that, to create room for transformative vision among stakeholders, it is key to support critical reflection and learning about transformations.

4.2 Public participation

Public participation concerns how citizens and stakeholders are involved in the policy-making process and is here used as equivalent to citizen participation or community participation that engage the public in decision-making processes. Different levels and forms of participation exist, as illustrated, e.g. in Sherry Arnstein (1969) well-known 'Ladder of participation' and in OECD's (2020) report on innovative citizen participation.

Public participation might, under certain circumstances, contribute to the legitimacy of less and slower mobility in regard to policy procedures. This could be the case if procedures allow and encourage people to participate, thereby potentially making the participants and/or other citizens consider the processes and procedures as more legitimate and by that also the adopted policy. The question of legitimacy as a consequence of participation is usually not answered directly in the literature, and the results regarding participation's contribution to sustainable outcomes within transport differ (Campbell et al., 2023; Rye et al., 2023). The crucial question with respect to the legitimacy of public participation involves the way that public participation is carried out. Thus, several criteria for public participation that contribute to legitimising sustainable outcomes can be drawn. Among important criteria are (1) to ensure representativeness and avoid majoritarianism (Cattino & Reckien, 2021; Pickering et al., 2022; Sørensen et al., 2023); (2) to invest in public awareness and education and thereby promote informed public debate (Pickering et al., 2022;Sørensen et al., 2023) and (3) to allow for dissensus in the participatory processes, since it might often be difficult to obtain consensus in contentious processes of sustainability transformations (Rye et al., 2023;Sørensen et al., 2023Sørensen et al., 2025).

A form of public participation that seems to play an increased role is representative and deliberative citizen participation, which OECD (2020) has considered to be a worldwide wave. This form of participation implies that a cross-representative sample of a given population is selected randomly across, e.g. locations, professions, ages, gender, class, car ownership, etc. and is thus broadly representative of a community. As the participants often meet in a number of sessions over several months, a deliberative approach implies that there is sufficient time to learn, listen, reason, discuss and suggest recommendations to policy makers. The approach has been applied in different climate citizen assemblies in many countries and cities (Giraudet et al., 2022; OECD 2020) and, in some cases, with significant results with respect to the recommendations provided (Sørensen & Isaksson, 2021).

4.2.1 Public participation for less and slower mobility

Representative citizen participation has also been applied in connection with the transport sector. In France, a citizen convention for climate with 150 citizens in June 2020 presented 149 concrete recommendations, 43 of which dealt with the transport sector. The recommendations included several measures that favour less and slower mobility. Thus, of all 43 recommendations dealing with transport, 24 can be characterised as *avoid* and *shift* approaches, e.g. restrictions and disadvantages for automobile and aeromobility, such as taxation of aeroplane fuel, removal of tax benefits for car use and reduced speed limits on highways, but also support alternative modes, such as introducing tax relief on train tickets. The support among the members of the convention for each recommendation was between 60% and 98%

(Sørensen & Isaksson, 2021). The example illustrates relatively high support of measures favouring less and slower mobility, though it should be stressed that measures intended to increase speed for (slower modes such as) bikes and public transport were also among the recommendations. At first, the political reaction was very positive, but later, the members of the convention were very critical with respect to the lack of political follow-up and implementation (Giraudet et al., 2022).

Another example is from the Danish capital, Copenhagen, where an assembly of 36 citizens was carried out in 2019 to suggest development plans for the city's medieval area. A total of 75% of the members of the citizen assembly were inhabitants in the area, while 25% came from outside. Among other things, the assembly recommended removing 750 of the 1,050 on-street parking slots in the area. The plan for this and other measures was to reduce car traffic by 45–60% and favour cycling and walking and the use of the space in the medieval area for other cultural and recreational activities. Consensus was not an aim of the assembly, and minority opinions were an option; however, ultimately, there were no minority options and compromises were found. The example illustrates considerable support for reducing car transport and favouring sustainable transport modes in the area, benefitting less and slower mobility. In the Copenhagen example, the City Council has adopted all the recommendations, including the removal of parking slots and allocated funding in the municipal budget, to start implementation (Portinson Hylander et al. submitted).

4.3 Experimentation

Experimentation (in the form of, e.g. living labs, testbeds, pilots and demonstrations) is a popular method for solving major societal changes related to sustainability issues, including mobility and transport (Sengers et al., 2019). In transition studies, experiments are seen as sites where new innovations can be tested in protected environments where users can become accustomed to sustainable products and services that they find meaningful without commercial pressure (van Waes et al., 2021). Experiments are also seen as sites for codesigning prototypes and learning about future everyday knowledge and practices (Sjöman & Hesselgren, 2022) and as a possibility to engage citizens as local experts (Günther et al., 2023; Henriksson et al. 2024b; Kesselring et al., 2023). Policy instruments aiming to build legitimacy in regard to the expected effects intend to strengthen citizens' beliefs in whether a policy can work or not and if policy actors have the capacity to set up the policy in question efficiently. Experimentation is a policy instrument that can contribute in particular to cognitive legitimacy concerning expected policy effects owing to its ability to illustrate, often on a small scale and in a very practical manner, that innovations, policies or other measures can work in practice. If citizens are made aware that a policy can work in their everyday life, they are likely to believe that they can work on a larger scale, or beyond the experiment, too.

4.3.1 Experimentation for less and slower mobility

We illustrate how experimentation can support legitimacy for less and slower mobility via two examples. The first is a larger scale pilot on congestion taxes that were introduced by public actors in Stockholm, Sweden, and the second is a smaller scale research led living lab aiming to reduce everyday car-based mobility.

The Stockholm congestion tax was introduced as a trial in Stockholm, Sweden, in 2006. The question of legitimacy was crucial since the Social Democratic party in the preceding municipal election campaign had promised *not* to implement a congestion charging scheme but was forced to do so, as the congestion tax became a key ingredient in negotiations following the elections when the party was dependent on support from the Green Party to gain a majority of seats in the local parliament. The decision to establish the congestion tax as a trial followed by a popular referendum was necessary to gain legitimacy (Isaksson & Richardsson, 2009; Sørensen et al., 2014). While only 33% of the citizens of Stockholm were in favour of the scheme prior to the trial (versus 56% opposing), a majority of 52% were in favour of the tax some months after (versus 40% opposing). In the referendum, 51.3% were in favour of the tax. However, outside of the city, residents remained critical. Some surrounding municipalities also arranged referenda, and on average, 60% were against and 40% in favour of congestion taxes (Sørensen et al., 2014).

The second example is Living Lab Riksten, which is a research-led living lab in Riksten, a newly established residential area on the outskirts of the Stockholm region in Sweden (Henriksson et al., 2024b).

A small group of participants were engaged as coresearchers in the living laboratory. They were asked to reduce their car trips and were given access to a number of accessibility and mobility services (e.g. a bicycle pool, e-scooters, and a telecommuting centre for remote work). In parallel, they were asked to reflect upon their mobility practices, local environment and sustainable policy in written reports. The results show that the coresearchers were not successful in reducing car trips, which reflects that their daily activities demanded car travel. The built environment and transport infrastructure, with limited access to sufficient public transport and walking and biking facilities, shaped their perception that they needed the car to support their everyday needs. Despite this, many people enjoy biking and public transportation, when possible, for example, for work commuting. Interestingly, the written reports revealed that the coresearchers were in favour of infrastructural investments and policies that supported sustainable mobility and made the car less attractive. The takeaway from the living lab is that when citizens are invited to try out new mobility services and suggest solutions for sustainable travel in their local environment, they are willing to support policies that would reduce car use, even at the expense of their own convenience.

5 Discussion

Having presented examples of catalytic policy instruments for less and slower mobility, we critically discuss their potential and preconditions for favouring less and slower mobility as well as their interlinkages and limitations. Table 1 shows how the examples are linked with policy aspects, policy legitimacy, catalytic policy instruments, catalytic potential and core preconditions for supporting legitimacy.

5.1 Potentials, preconditions and limitations

The first catalytic policy instrument analysed was visioning. Considering the insights from two European examples, we propose that visioning can strengthen norms and values that favour less and slower mobility and thus the policy preconditions through opening discussions of how citizens imagine that they want to live. There are, however, challenges with visioning exercises. One issue is how to open up discussions where the values and conflicts of different futures can be debated rather than silenced in the striving for futures that are attractive and argued legitimate by different groups despite their different interests and

Table 1. Three catalytic policy instruments, related to policy aspects, policy legitimacy dimension, examples applied, catalytic potential and preconditions for being catalytic.

Policy aspect relevance	Policy legitimacy dimension	Catalytic policy instrument	Examples applied	Potential of PI catalysing legitimacy for less and slower mobility	Core preconditions for realizing catalytic potential
Policy preconditions	Normative	Visioning	- Four scenarios for mobility in Europe - Co-production of transformative and contextual visions in Europe	Visioning may strengthen normative legitimacy and thus legitimacy of policy preconditions for less and slower mobility by inviting citizens to imagine how they want to live sustainable lives.	That visions of less and slower mobility include issues of liveability and wellbeing.
Policy procedures	Input/ Procedural	Public participation	- French citizen convention for climate - Copenhagen citizens' assembly	Public participation may contribute to legitimacy of policy procedures by grounding recommendations of less and slower mobility among the broader segments of the population.	- Observe representativeness, public awareness and education, and allow for dissensus. - Media coverage of participatory processes to inform the broader population.
Policy effects	Output/ substantive/ cognitive	Experimentation	- Stockholm congestion charging - Living Lab Riksten	Experimentation may contribute to cognitive legitimacy and thus legitimacy of expected policy effects by providing citizens with concrete experience of consequences of changes in their local environment and everyday lives.	- Experiments involve local stakeholders and authorities that can adopt results and methods in ordinary planning processes. - Engage citizens that are willing to try new solution that are adopted to everyday needs in the experiments.

norms (Gunnarsson-Östling, 2014). Sustar et al. (2020) argue that methods to enable critical discussions on different futures are lacking. Furthermore, it is difficult to assess in what way positive visions contribute to transformations in practice (Nikoleris, 2018).

Even if it is difficult to prove that visioning exercises lead to legitimacy for actual policy interventions, the insights from two European examples indicate that if positive future visions of a life with less and slower mobility are discussed widely in society, it has the potential to change norms, values, and attitudes. Widening the discussion of how we imagine that we want to live, with a focus on livability rather than transport per se, can be key for strengthening the legitimacy of less and slower mobility in the policy preconditions. It might be that a vision of zero emissions—or less and slow mobility—in itself does not engage people but rather a vibrant local society and well-being does (Cameron & Potvin, 2016). The examples illustrate that both citizens and policy makers can constructively be involved in exercises that focus on how to envision a more sustainable future and thus support the legitimacy of policy preconditions. If visions for a life with less and slower mobility is discussed in terms of wellbeing and in relation to equity, widely in society, it might have potential for changing norms, values, and attitudes in the population.

The second instrument discussed was public participation, where our examples dealt with representative and deliberative participatory processes. In both examples, the citizen assemblies contributed to legitimacy in the broader population for measures of less and slower mobility only to a limited extent, while in the Copenhagen example, the citizen assembly has provided legitimacy for measures mostly vis-à-vis the political authorities. This form of representative and deliberative participation seems to honour most of the criteria mentioned above (section 4.2) regarding representativeness and majoritarianism, awareness and education, as well as the issue of consensus, since dissensus was allowed, though in the Copenhagen assembly, the recommendations had support from all members.

Representative and deliberative participatory processes may contribute to legitimacy concerning policy procedures by grounding recommendations of less and slower mobility among the wider population. However, it requires a greater degree of media coverage of the assemblies' aims, roles, discussions and recommendations, which is also a conclusion from a Scottish citizen assembly (Elstub et al., 2022). Whether organised as citizen assemblies or not, public participation following the criteria mentioned in Section 4.2 might contribute to legitimizing policies for less and slower mobility, but more should be done to provide for the assemblies contributing to the legitimacy of procedures of such policies in the wider population.

Finally, experimentation as a catalytic policy instrument may contribute to the legitimacy of anticipated policy effects. Based on the examples provided as well as the literature, experimentation provides citizens with important experience with policy effects in their everyday lives. Like challenges with vision as a catalytic policy instrument, critics claim that it is difficult to assess whether experimentation is successful in the diffusion (upscaling) of sustainable innovations (von Wirth et al., 2019). Another critical issue that has been raised is in what way lessons and take-aways from experiments, especially those that are initiated by researchers or commercial partners, are linked back to and integrated into conventional planning processes (Oldbury et al., 2022; van Waes et al., 2021). On the other hand, there are some persuasive arguments on the merits of experimentation in the arena of mobility. In a German study of three different mobility living labs, Kesselring et al. (2023) emphasized that experiments aimed at changing mobility systems can significantly increase their impact when planned in cooperation with citizens and other stakeholders. When activating citizens and their practical knowledge, they are likely to become ambassadors for novel solutions or policies since, through the experiment, they know that the changes will work and that they solve problems they experience in their local contexts (ibid.). To bring forth practical experiences is, we argue, the key point of cognitive policy tools. Moreover, the potential of different solutions, whether they are geared towards new technology or towards policy measures such as taxes or subsidies, risk failure if the present environment does not support sustainable mobility. The fact that the experiment is aligned with current goals and objectives as well as future plans is also a pressing issue for success, as we observed in the Stockholm congestion tax example. To that end, engaging public actors and other local stakeholders with the mandate to influence the built environment in experimentation is probably as vital as engaging citizens (Oldbury & Henriksson, submitted).

Overall, the catalytic policy instruments we have introduced in the article are *potentially* catalytic, as such instruments, e.g. liveability and wellbeing (visioning), media coverage and information (public participation), and integration with local policies (experimentation), complement the use or are a result of the policy instruments.

5.2 Interlinkages

Catalytic policy instruments may be linked. Visions can be developed via citizen participation, as in the case of four scenarios for mobility in Europe (Section 4.1) and Copenhagen (Section 4.2). Experiments can be aligned with and take place within an image of an envisioned future, as was the case with the trial on congestion charging in Stockholm (Section 4.3); similarly, visions of the future can include increased democracy and participation. Linking the instruments in such ways might even strengthen the catalytic capacity of each instrument to contribute to its legitimacy. The merging of more of these instruments in a process likely increases the legitimization effect of each. Policy packages (Givoni, 2014), which we see as another catalytic policy instrument, can link more of these instruments with other transformative policy instruments, and the process of establishing a package can include elements of public participation.

Furthermore, it is imperative to stress that future visions, including aspects of justice, fairness and equity, can support legitimacy and thus have stronger catalytic potential. While both effectiveness and equity are typically considered important for policy acceptance (Banister et al., 2007), perceptions of justice and fairness have been identified as the most important factor in public opinions about climate change policies, followed by the mitigation effectiveness of policies (Bergquist et al., 2022). This means that, to secure legitimacy for policies, policy makers should aim to address justice-related issues. However, what is considered fair or just depends on scale and context (e.g. within or between cities, regions and nations, cf. Mattioli, 2016) and therefore might be a difficult guideline for policy makers aiming for the legitimacy of less and slower mobility. Moreover, justice can also be mobilized to obstruct meaningful sustainability policies (Ekberg et al., 2022; Lamb et al., 2020).

Finally, catalytic policy instruments may weaken each other. All the suggested catalytic policy instruments are time-consuming and involve the risk of backlash and unanticipated consequences. Many examples exist of citizens not favouring policies of less and slower mobility, and we know of protests regarding road tolls, congestion charging or increased taxes on fuel (Wanvik & Haarstad, 2021). Thus, there is a risk that participation reduces the legitimacy of transformative policies and contributes to the rejection of visions and experiments aimed at transformation (Rye et al., 2023). Building public legitimacy for less and slower mobility in a society based on and idealising high mobility is thus an arduous task, and democratic processes also involve the risk of failure, but this is perhaps a price worth paying to build and uphold legitimacy for substantial transformations.

6 Conclusions

The aim of this article is conceptual development by suggesting *catalytic policy instruments* as tools that can help gain legitimacy for less and slower mobility. Catalytic policy instruments are defined by their potential to substantially transform conditions and shift the groundwork for future policy making, thus providing conditions for introducing other transformative policy instruments. The catalytic potential of a certain instrument lies in shifting the level of a policy's legitimacy either in an ongoing or subsequent policy process.

We analysed three catalytic policy instruments: visioning, public participation and experimentation. They may contribute to gaining legitimacy for less and slower mobility in three distinct ways: by changing norms and values and thereby establishing the legitimacy of policy preconditions (visioning), through policy procedures that legitimize policies for less and slower mobility (public participation) or by using experiments to make anticipated policy effects more legitimate (experimentation). Even when backlashes are experienced, we expect the consequences with respect to legitimacy to be reduced compared to a situation without the use of catalytic policy instruments (cf. Figure 2). Localised visioning exercises, public participation processes and experimentation carefully and adequately applied may also help find solutions for less and slower mobility in difficult contexts, such as rural or peripheral areas, or concerning citizens

with poor accessibility (e.g. disability, low-income or care responsibilities). In such cases, changes in the mobility system must also closely connect to changes in the built environment and localising of societal services (Lyons et al., 2024).

We have argued that what is considered in accordance with attitudes, norms and values and what is considered legitimate procedures and effects differ between countries and across time. The examples in this article all stem from a European context, and it is likely that examples from other countries and regions will provide other insights into experiences with visioning, public participation and experimentation as instruments to provide legitimacy for less and slower mobility. Cautiousness concerning transferability to other countries and context should therefore be observed. In addition, the employment of catalytic policy instruments to gain legitimacy for less and slower mobility is no Egg of Columbus. Catalytic policy instruments should be applied strategically and with contemplation, and even then, the instruments will compete with many other influential processes and trends in the transport sector and wider society that shape our understanding of what are considered legitimate policies. Even if policies of less and slower mobility in broad segments of the population were to gain legitimacy, there would still be strong actors striving to maintain the status quo because of vested interests in the political economy of hypermobility.

To conclude, there is a need for more research on catalytic policy instruments for less and slower mobility. There is a need for more cases from other countries and cultural contexts, and other potentially catalytic policy instruments should also be studied to understand how these instruments work and contribute to the establishment of conditions in favour of less and slower mobility, which could contribute to a more general understanding of the catalytic potential of policy instruments for less and slower mobility. Future studies should thus focus on including other data sources, such as time horizons, metrics on pre/post support levels, voting results and the durability of the effects of policy instrument application. We also need more pilots on public participation, more positive future visions of life with less and slower mobility, as well as experiments testing such policies. With more studies and literature on catalytic policy instruments for less and slower mobility, it would also be possible to develop actionable guidelines for policy makers, planners, civil society organisations and others engaged in the transformation of the sector.

Endnotes

1. See www.cittaslow.org
2. See https://westayonthe ground.org

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